MIL-F-16081G

10 June 1971
SUPERSEDING
MIL-F-0016081F(GL)
1 February 1971 and
MIL-F-16081E
18 April 1968

MILITARY SPECIFICATION

FANS, VENTILATING, PROPELLER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers propeller-type ventilating fans for horizontal or vertical mounting.

* 1.2 Classification.- Fans shall be of the following types, classes, styles, and sizes (diameters) as specified (see 6.2).

Type I - Nonhazardous location

Class 1 - Direct drive

Style A - Wall or window mounting

Sizes - 12, 16, 18, 20, 22, 24, and 30 inch (nominal propeller diameter)

Style B - Roof or ceiling mounting

Sizes - 16 and 24 inch (nominal propeller diameter)

Class 2 - Belt drive

Style B - Roof or ceiling mounting

Sizes - 24 and 48 inch (nominal propeller diameter)

FSC 4140

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Type II - Hazardous location

Class 1 - Direct drive

Style A - Wall or window mounting

Sizes - 12 and 18 inch (nominal propeller diameter)

Style B - Roof or ceiling mounting

Sizes - 16 and 24 inch (nominal propeller diameter)

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/1	-	Aluminum 1100, Plate and Sheet
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
TT-P-664	-	Primer, Coating, Synthetic, Rust-Inhibiting, Laquer-Resisting
PPP-B-601	-	Boxes, Wood, Cleated Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	-	Box, Fiberboard
PPP-C-650	-	Crates, Wood, Open and Covered

MILITARY

MIL-P-116	-	Preservation, Methods Of
MIL-T-152	-	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic and Associated Electrical Equipment
MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

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STANDARDS

FEDERAL

FED-STD-595 - Colors

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-147 - Palletized and Containerized Loads 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet Type Base
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of

(Copies of specifications, standards, and publications, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

Standard No. 62 - Flexible Cord and Fixture Wire
Standard No. 498 - Attachment Plugs and Receptacles
Standard No. 508 - Industrial Control Equipment
Standard No. 698 - Industrial Control Equipment for use in Hazardous Locations

Hazardous Location Equipment List

(Application for copies of the Underwriters' Laboratories, Inc. (UL) publications should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; Walt Whitman Road, Melville, Long Island, N.Y. 11749; or 1655 Scott Boulevard, Santa Clara, California 95050.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION STANDARDS (NEMA)

Motors and Generators MG1

(Application for copies of NEMA Standards should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N.Y. 10017.)

RUBBER MANUFACTURERS ASSOCIATION, INC. (RMA)

Engineering Standards for Multiple V-Belt Drives

(Application for copies should be addressed to the Rubber Manufacturers Association, Inc, 444 Madison Avenue, New York, New York 10022.)

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Standard No. 210 - Test Code for Air Moving Devices
Standard No. 300 - Test Code for Sound Rating Air Moving Devices
Standard No. 301 - Method of Publishing Sound Ratings
Standard No. 302 - Application of Sonic Loudness Ratings for NON DUCTED Air Moving Devices

(Application for copies should be addressed to the Air Moving and Conditioning Association, 30 West University Drive, Arlington Heights, Illinois 60004.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code
Standard No. 91 - Blowers and Exhausts
Standard No. 96 - Ventilation of Restaurant Cooking Equipment

(Application for copies NFPA Standards should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, Massachusetts 02110.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article, inspection and approval (see 4.2, 6.2 and 6.3).

3.2 Standard product.- The fans covered by this specification shall be a regular commercial product of the manufacturer, except for any changes necessary to comply with the specification requirements. All parts shall be new and all fans or components furnished under any one contract shall be identical.

* 3.3 Codes and standards.- The fans shall comply with the following:

(a) The cords and wires shall conform to UL Standard No. 62. The plugs and receptacles to UL Standard No. 498, the controls for type I fans to UL Standard No. 508 and for type II fans to UL Standard No. 698.

(b) The motors as applicable shall conform to Standard MG1 of the National Electrical Manufacturers Association (NEMA)

(c) All fans shall conform to NFPA Standard No. 70. In addition type II fans shall conform to NFPA Standards No. 70 and No. 91.

(d) Fans shall conform to Standards No. 210, No. 300, No. 301 and No. 302 of the Air Moving and Conditioning Association, (AMCA).

* 3.3.1 Certification.- Prior to approval of the first article, or if none is submitted, prior to approval of the first shipment; the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the fans he proposes to furnish under this specification meet the requirements specified in 3.3.

3.3.1.1 NEMA and NFPA.- Acceptable evidence of meeting the requirements of NEMA Standards No. MG1, NFPA No. 70 and NFPA 91 shall be the manufacturer's certified statement that the fan(s) conform(s) to the applicable requirements.

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* 3.3.1.2 UL.- Acceptable evidence of meeting requirements of 3.3(a) shall be the UL label or listing mark or a certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the fan cords, wire, plugs, receptacles, fittings and controls conform to applicable UL standards.

* 3.3.1.3 AMCA.- Acceptable evidence of meeting the requirements of AMCA Standards 210, 300, 301 and 302 as applicable shall be the AMCA Certified Rating Seal, or a certified inspection report from an independent testing laboratory indicating that the fan conforms to the requirements of AMCA Standards 210, 300, 301 and 302 as applicable.

* 3.4 Materials.- When materials are not definitely specified, they shall be of the quality normally used for propeller ventilating fans in commercial practice, provided the completed item complies with all provisions of this specification.

3.5 Design.- The exhaust ventilating fan shall consist essentially of a propeller, inlet venturi, mounting frame, electric motor, motor support frame and accessories as specified herein. The motor and motor support frames shall be attached to the mounting frame so that adequate clearance between the propeller and venturi shall be insured. The fan shall be designed so that it will comply with requirements hereinafter specified. The design shall be such that it will permit accessibility for field maintenance.

* 3.5.1 Fan characteristics.- Unless otherwise specified (see 6.2), the fans shall have the characteristics listed in table I and shall conform to the AMCA Standard Codes 210, 300, 301 and 302 when tested as specified in 4.4.1. For special applications whenever more precise determination or sound levels in installed applications are required (see 6.5).

TABLE I.- Fan characteristics

* Class	Nominal Size	Nominal capacity C.F.M. @ zero inches water
1	12	1000
	16	1600
	18	2500
	20	3200
	22	3200
	24	4700
	30	6600
2	24	6800
	48	22000

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3.6 Details of construction.-

3.6.1 Mounting frame and inlet venturi.- The mounting frame and inlet venturi may be of either one piece or separate construction. In either case the mounting frame shall consist of a rigidly constructed metal frame designed to provide support for the motor, inlet venturi, and other components of the fan such as wall, ceiling, window, or roof mounting, as applicable. Mounting holes shall be provided in the frame for complete fan installation. The inlet shall be of a configuration which will provide quiet, efficient air flow and meet the air flow requirements of table I.

3.6.2 Propeller wheel.- The propeller shall have two or more blades which shall be statically balanced and shall operate as intended. The propeller shall be secured to the driving shaft in such a manner as to insure positive rotation and easy removal of the propeller from the shaft. The propeller shaft for class 2 fans shall be separately mounted from the motor and the two shall be connected by a V-belt and pulleys, to provide performance as specified herein. A device shall be incorporated to permit adjustment of V-belt tension for class 2 fans. All pulleys shall be secured to the respective shafts in a manner which provides for positive rotation and easy removal for maintenance. The blades for type I fans shall be fabricated of a corrosion-resistant material or treated to resist corrosion. The blades for type II fans shall be fabricated of nonferrous material treated to resist corrosion. Blades shall be rigid enough to withstand normal usage and cleaning without bending out of alignment.

* 3.6.3 Motor.- The motor for type I fans shall be totally inclosed of the fan duty type. The motor for type II fans shall be the explosion proof type for operation in hazardous locations as defined by NFPA 70, 91 or 96 as applicable. The motor shall conform to NEMA STD MG-1. The motor shall be for continuous fan operation for the voltage, frequency, and phase characteristics specified (see 6.2).

* 3.6.3.1 Motor mounting.- For class 2 fans, the motor shall be mounted either inside or outside the air stream as specified (see 6.2). When specified (see 6.2), motors located outside the air stream shall have a V-belt drive totally inclosed in a housing for protecting the V-belt drive from deterioration. Support for the motor and propeller wheel shall be mounted on the air inlet side of the fan and shall be adequately braced so that during fan operation, alignment of the motor and propeller wheel assembly with the air inlet opening shall be maintained. V-belt drives shall conform to RMA Engineering Standards for Multiple V-Belt Drives and have a service factor of 1.3.

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* 3.6.3.2 Motor input.- The current input, at rated voltage, shall not exceed the motor plate current rating by more than 15 percent for any air volume output between the rated capacity at zero inch water gage static pressure and the rated capacity at 1/4 inch water gage static pressure. The motor input shall be determined during fan performance test as specified in 4.4.1.

* 3.6.3.3 Bearings.- All bearings, except for style A fans, shall be standard, commercial, sealed-in lubrication, antifriction type. For style A fans, commercial sleeve bearings may be used in lieu of the sealed-in lubrication antifriction bearings. Bearings shall be adequate for both radial and thrust loads occuring in the vertical or horizontal mountings of the wall and roof fans.

3.6.4 Power connection.- The fans shall either be furnished with a three-conductor neoprene-covered flexible cable (type SO) and a three-prong attachment plug (see 3.6.4.1) or be equipped to be wired into the line as a permanent installation (see 3.6.4.2) as specified (see 6.2). All power or electrical connections shall meet the requirements of the National Electrical Code.

* 3.6.4.1 Cable and attachment plug.- All fans for use only on nominal 115-volt, single phase operation shall be furnished with cable and plug. The cable shall not be less than 6 feet long, color-coded, (black, white, and green). The conductors of the cable shall have the capacity to carry at least twice the current rating designated on the motor name plate and shall not be less than No. 18 AWG. One end of the green conductor shall be connected to the grounding contact of the attachment plug and the other end of the conductor shall be attached to the frame of the fan motor. Strain-relief devices shall be used at both the fan and plug end of the cable for security.

3.6.4.2 Permanent installation.- When permanently installed, the fan shall be provided with means for the connection of armored cable or conduit and shall have a terminal box so located that after installation it shall be readily accessible for field-wiring and inspection. The terminal compartment shall be so attached as to be prevented from turning with respect to the fan. The conduit opening in a roof-mounted appliance shall be threaded unless (1) it is located below the lowest uninsulated live part within the enclosures, or (2) its location prevents drainage into the enclosure. The conduit opening in type II fans shall be threaded. The permanently installed fan shall be provided with wiring terminals for the connection of conductors suitable for the fan in accordance with the National Electrical Code, or it shall be provided with appropriate leads for such connection.

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* 3.6.4.3 Fuse size or time limit circuit breaker marking.- When ventilating fans are not equipped with thermal cutouts, relays, or other devices capable of opening short circuits, the equipment nameplate shall be marked to indicate the maximum size of fuse or time limit circuit breaker that is to be used to protect the equipment in accordance with the National Electrical Code (NFPA No. 70).

3.7 Accessories.- The following accessories shall be furnished, when specified (see 6.2).

3.7.1 Manual starter.- The starter shall be designed for manual operation and shall meet the requirements of UL Standard 508. The starter shall be a NEMA type and of the proper size and capacity for the fan motor, as required by the National Electrical Code (see 6.2).

* 3.7.1.1 Overload protection.- The circuit shall be disconnected automatically when overloaded, by means of a thermal overload device set up to trip as required by the National Electrical Code. A manual or automatic reset, as specified (see 6.2), shall be provided. The manual starter shall be designed to prevent recycling when the operating handle is held in the ON position, and the motor is subjected to an overload sufficient to trip the starter or when a short-circuit occurs.

3.7.1.2 Starter enclosure.-

3.7.1.2.1 Type I.- The starter enclosure for the type I fan shall be a NEMA type I with knockouts for conduit connections at top and bottom of enclosure.

* 3.7.1.2.2 Type II.- The starter enclosure for type II fans shall be a hazardous locations type as defined by NFPA 70, 91 and 96 as applicable. Unless otherwise specified (see 6.2), a 3/4-inch conduit fitting hole shall be provided.

3.7.2 Propeller guard.- The guard shall be constructed of metal in such a manner as to permit ready attachment and removal on the air inlet side of the fan frame with fastening devices provided for secure installation. When installed, the guard shall provide for protection to personnel and to the propeller blade without affecting the capacity or efficiency of the fan, as outlined in table I. Propeller fan guards shall be furnished on installations less than 7 feet above the floor (see 6.2).

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3.7.3 Gravity shutter (style A).- The gravity shutter shall have a square or rectangular corrosion-resistant metal frame of proper size to suit the fan diameter. Mounting holes shall be provided around the periphery of the frame to permit a secure and complete installation of the equipment at assembly. The gravity shutter shall be provided with louvers to cover the frame opening. The louvers shall be constructed of 1100 or 3003 aluminum of any suitable temper conforming to QQ-A-250/1 and QQ-A-250/2, respectively. The louvers shall be counterbalanced to open fully and freely by the action of the discharged air with minimum flutter and minimum resistance to air flow. When the fan is idle, the louvers shall close automatically and shall cover the opening completely. A means of interconnecting the louvers shall be provided so that all louvers operate as a unit. All hinge parts shall be corrosion-resistant.

3.7.4 Housing (style B).- The style B fan assembly shall be securely mounted in either a penthouse (fig. 1) or a hood-type housing (fig. 2) as specified (see 6.2). Penthouse-type housing shall be provided with gravity shutters as specified in 3.7.3. Hood-type housing shall be provided with automatic opening and closing louvers. The base of the housing shall be designed to fit over a square, rectangular or round curbing and the inside dimensions of the base and the maximum housing height shall be as specified (see 6.2 and figures 1 and 2). The housing shall be capable of withstanding the test specified in 4.4.4 without distortion of the housing or any of its components. The housing shall be hinged to permit ready access to the fan assembly and other components for maintenance purposes. Locking devices shall be provided for locking the housing in the closed position. All seams and joints shall be lightproof and weatherproof. The housing shall be fabricated of corrosion resistant metal or steel, treated to resist corrosion.

3.7.4.1 Metal thickness.-

Hood	- Not less than 0.0359 inch (nominal)
Airshaft	- Not less than 0.0478 inch (nominal)
Base	- Not less than 0.0598 inch (nominal)

3.7.4.2 Blackout properties.- When specified (see 6.2), the design of the housing shall be such that any exposed area of the ventilator when viewed from any point, on a horizontal plane, level with the roof line and from any point vertically away from the horizontal plane, through 180 degrees in any direction, shall not show light leakage.

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3.7.5 Screen.- The screen shall be made of wire mesh of corrosion-resistant metal or steel, treated to resist corrosion. The wire mesh provided shall be secured to the housing in such a manner as to prevent the entry of birds, or other foreign objects into the housing interior. Unless otherwise specified (see 6.2), the screen mesh shall be not less than 1/2 inch by 1/2 inch.

* 3.8 Moisture and fungus resistance treatment.- When specified (see 6.2), all the circuit electric elements shall be treated to resist fungus and moisture as specified in MIL-T-152. Circuit elements shall include, but shall not be limited to cable, wire, switchboards, panel boards, terminal and junction blocks, junction boxes, and coils and capacitors.

3.9 Instruction plate.- Each fan shall be equipped with instruction plate, suitably located, describing any special or important procedure to be followed in operating and servicing the equipment. The instruction plate shall be aluminum, copper, or zinc base alloy.

3.10 Nameplate.- The manufacturer's standard nameplate shall be permanently affixed to the fan in a comspicuous place.

3.11 Marking for identification.- The fan shall be marked for identification in accordance with the marking requirements of MIL-STD-130.

3.12 Lubrication information.- The supplier shall affix to each item of equipment a lubrication chart (metal or decal) indicating the type and grade lubricant required and the time interval when lubrication should be accomplished. The lubrication instructions and printed lubrication chart shall also be included in the equipment manual.

* 3.13 Electromagnetic compatibility.- When specified (see 6.2), equipment shall be designed and equipped for electromagnetic compatibility in accordance with class II B of MIL-STD-461 (see 4.4.3).

3.14 Finish.- Unless otherwise specified (see 6.2), the equipment, including electrical components not required to be fungus proofed, shall be treated and painted with a commercial paint in accordance with the manufacturer's standard commercial practice. The color shall conform to No. 16187 gray in accordance with FED-STD-595. There shall be no blisters, peels, chips, or areas of thin film or no film.

3.15 Workmanship.-

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3.15.1 Steel fabrication.- Steel used in the fabrication of equipment shall have no kinks or sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing shall be done neatly and accurately. Corners shall be square and true. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape.

3.15.2 Bolted connections.- Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with commercial practice, and all bolts, nuts, and screws shall be tight.

3.15.3 Riveted connections.- Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools, and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.15.4 Welding and brazing.- All surfaces of parts to be welded or brazed shall be clean. Welding and brazing shall be accomplished in a manner which will prevent the occurrence of burn holes, cracks, fractures or incomplete fusion. All scale or flux deposits shall be removed from finished welds.

3.15.5 Electrical connections.- Electrical connections shall be soldered or brazed, except connections on screw type terminals shall be mechanically secured. For soldering electrical connections, resin based fluxes shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a first article is required, it shall be examined in accordance with table II, specified dimensions, and tested as specified in 4.4.1, 4.4.2, 4.4.3, 4.4.4, and 4.4.5, as applicable. The presence of any visual defect, any dimension not within the specified requirements or failure to pass all tests shall be cause for rejection of the first article.

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* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The lot shall consist of all completely finished fans of one type, class, style and size offered for inspection at one time. The sample unit for this inspection shall be one completely assembled finished fan.

4.3.2.1 Visual examination.- Examination of the end item shall be in accordance with the classification of defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 total defects expressed in terms of defects per hundred units.

TABLE II.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Finish and color not as specified	X	
	Blistered, peeled, chipped, or area of no film or thin film		X
	Not finished where required	X	
Design	Characteristic not in accordance with the specified requirements	X	
Construction and workmanship general	Part missing or not specified type or size	X	
	Fractured, split, punctured, dented, deteriorated, bowed, or malformed	X	
	Misplaced, loose or not in proper alignment	X	
	Sharp burr, sliver or splinter	X	
	Operation omitted or not properly performed	X	
	Functioning component that require abnormal force to operate		X
	Not connected or joined as specified or poorly accomplished	X	
	Adjustable assembly that cannot be adjusted	X	
	Adjustable assembly not properly adjusted to perform the function intended		X

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TABLE II.- Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Welding and brazing	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Bolts, nuts, screws, studs and other types of threaded fasteners	Missing, not typed specified, broken, stripped, fractured, or loose		X
Assembly	Component not properly assembled or secured	X	
	Unit perceptibly out of square or alignment	X	
Marking, identification, instruction plate, and lubrication chart	Missing, incomplete, not legible		X

4.3.2.2 Dimensional examination.- Inspection shall be made of fans for compliance with dimensions specified. Any dimension that is not within the specified tolerances shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

* 4.3.2.3 Testing of end item.- When a first article is not required, the initial production unit shall be tested as specified in 4.4.1, 4.4.3, 4.4.4, and 4.4.5, unless otherwise specified (see 6.3). Each unit of production shall be tested as specified in 4.4.2.

4.3.2.4 Standards and publications compliance.- Proof of compliance with the requirements of 3.3 shall be available to the contracting officer or his representative.

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4.3.2.5 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table III. The sample unit shall be one shipping container fully prepared for delivery, except that the container need not be sealed. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 with an acceptable quality level of 4.0 defects expressed in terms of defects per hundred units.

TABLE III.- Examination of preparation for delivery

Examine	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservative improperly applied or missing
Materials	Component missing or damaged
Workmanship	Bulged or distorted containers
Lubrication chart	Missing

4.3.2.6 Examination of shipping containers.- When shipping containers are required in accordance with PPP-B-636, the examination for defects in closure, waterproofing and strapping shall be in accordance with that specification.

4.4 Tests.-

* 4.4.1 Fan performance.- The fans furnished under this specification shall be connected to power and operational tested to determine compliance with 3.5.1 and 3.6.3.2. The test shall be in accordance with AMCA Standards 210, 300, 301 and 302. All results shall be corrected to standard air conditions (see 6.4). In lieu of this performance test an AMCA certificate (see 3.3.1.3), may be presented.

* 4.4.2 Production tests.- The fan shall be tested to withstand a high potential in accordance with NEMA MG1. When connected to the rated voltage, the current input shall not exceed the nameplate rating in amperes. During this operational test, the fan shall be examined for malfunctions. Failure to pass the above tests shall be cause for rejection of the unit.

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* 4.4.3 Electromagnetic compatibility.- When electromagnetic compatibility is required, the first article or initial unit of production, as applicable, shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461.

* 4.4.4 Housing wind test.- The housing shall be firmly attached at the base and subjected to a wind force of not less than 80 m.p.h. on all sides, respectively, for not less than 15 minutes per side. Examination shall be made to determine compliance with 3.7.4. Any nonconformance shall constitute failure of this test.

* 4.4.5 Blackout test.- When blackout properties are required, the test shall be made in a dark room free from any light infiltration. The completely assembled ventilator unit shall be mounted on a hollow base containing a light source, directly over a horizontal window equivalent in size to the base of the ventilator. This window shall be made of a light-diffusing material such as opal glass with edges sealed into the opening in the top of the hollow base. The joint between the hollow base and the bottom of the ventilator shall be fitted with a lightproof seal. The light-source window shall be illuminated from within the hollow base with two 100 watt bulbs not more than 12 inches from the window. The fan wheel shall be stationary and the shutter fixed in its wide-open position during the test. A horizontal roof deck constructed in normal relationship to the actual roof surface surrounding the ventilator unit when in place shall be provided. The horizontal deck shall extend not less than 18 inches beyond the outer limits of the ventilator and shall be made of a continuously flat surface material. Observations shall be made to determine compliance with 3.7.4.2. Any nonconformance shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Disassembly.- Disassembly shall be the minimum necessary to safeguard parts known to be especially subject to damage or loss and to accomplish a reduction in cube. Disassembly shall be such that no special skills and special tools are required to place the equipment in operation. Attaching hardware shall be placed in a cotton cloth drawstring bag and the bag closed and secured to one of the mating parts.

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5.1.1.2 Preservation.- All preservation shall be in accordance with material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.2.1 Cleaning.- All unpainted, uncoated, exposed ferrous metal surfaces of each fan, including detached parts and accessories, as required except corrosion-resisting steel surfaces, shall be cleaned process C-1 and thoroughly dried.

* 5.1.1.2.2 Class 1 fans.- Each fan, including accessories when attached, shall be preserved method IA-15 or IIe as specified (see 6.2).

5.1.1.2.3 Class 2 fans.- All cleaned surfaces of each fan, except ferrous metal shafts and pulleys, shall be coated with type P-1 preservative. All cleaned surfaces of ferrous metal shafts and pulleys shall be coated with primer conforming to TT-P-664, the primer dried and belts replaced.

5.1.1.3 Packaging.-

5.1.1.3.1 Class 2 fans.- Movable components of each fan shall be secured in a fixed position in a manner as to prevent movement or damage during transit. The power cord shall be compactly coiled and the coil secured to the fan with jute, sisal, manila, or cotton cord or twine. Drive belts shall have the tension relieved.

* 5.1.1.3.2 Accessories.- Loose accessories (including housing for style B fans, when furnished knocked down) and small disassembled parts for each class 1 or class 2 fan shall be packaged in one or more snug-fitting fiberboard boxes conforming to type CF or SF, class domestic; variety and style optional of PPP-B-636. Movable components shall be secured as specified in 5.1.1.3.1. Contents of fiberboard boxes shall be immobilized and protected against physical or mechanical damage by means of fiberboard inserts made of corrugated fiberboard as specified for the boxes or other suitable cushioning material. Boxes shall be closed in accordance with the appendix of the box specification.

* 5.1.1.3.3 Class 2, style B fans completely assembled.- Fans, when furnished completely assembled, shall have movable components of the fan and accessories secured as specified in 5.1.1.3.1. Maintenance tools for each fan shall be packaged in one or more snug-fitting fiberboard boxes conforming to type CF or SF, class domestic, variety and style optional of PPP-B-636. Each box shall be closed and waterproofed by means of tape in accordance with the appendix of the box specification. The packaged maintenance tools shall then be placed in a void in the housing and secured to prevent movement during transit and storage. Access doors shall then be locked in the closed position by means of locking devices provided thereon.

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5.1.2 Level C.- Fans shall be preserved and packaged to afford adequate protection against deteriorating corrosion and physical damage during shipment from the supplier to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.-

* 5.2.1.1 Fans together with accessories exceeding 65 pounds.- Each fan together with equipment exceeding 65 pounds (except class 2, style B fans when completely assembled), preserved and packaged as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to overseas type, type 3 load of PPP-B-601 or class 2, style 2, 4, or 7, type 3 load of PPP-B-621. Shipping containers conforming to PPP-B-601 and style 2 or 4 of PPP-B-621 exceeding 250 pounds gross weight shall be provided with skids as specified in accordance with the appendix of the applicable specification. Each shipping container except those with contents preserved as specified in 5.1.1.2.2 shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Contents of each shipping container shall be secured by means of lumber blocking and bracing with cushioning used between the lumber blocking and unprotected finished surfaces. The blocking and bracing shall not depend on end-grain nailing, but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

* 5.2.1.2 Class 2, style B fans completely assembled.- Each fan completely assembled, preserved and packaged as specified in 5.1, shall be packed upright in a crate conforming to type IV or V of PPP-C-650. Type V crates shall be used when overall dimensions of the fans exceed the dimensional limitations of style IV crates. The width dimensional limitation for style V crates may be waived. Boxes containing maintenance tools shall be as specified in 5.1.1.3.3. The fan shall be anchored within the crate by means of bolting and the crate closed and strapped as specified in the appendix of the crate specification.

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* 5.2.1.3 Fans together with accessories up to 65 pounds.- Each fan together with equipment weighing up to 65 pounds, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s style optional of PPP-B-636. Contents shall be immobilized within the container by means of fiberboard inserts made of fiberboard conforming to type CF of PPP-B-636, having a minimum bursting strength of 200 p.s.i. Each shipping container shall be closed, waterproofed by means of tape and reinforced with flat strapping or tape banding in accordance with the appendix of the container specification. Waterproofing of shipping containers shall not be required when contents are preserved as specified in 5.1.1.2.2.

5.2.2 Level B.-

* 5.2.2.1 Fans together with accessories exceeding 65 pounds.- Each fan together with equipment exceeding 65 pounds (except class 2, style B fans when completely assembled, preserved and packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that domestic type or class shipping containers shall be used when weight permits and case liners shall not be required. Strapping of nailed-wood shipping containers shall be in accordance with the class 2 requirements of PPP-B-621.

* 5.2.2.2 Class 2, style B fans completely assembled.- Each fan completely assembled, preserved, and packed as specified in 5.1, shall be packed as specified in 5.2.1.2. Boxes containing maintenance tools shall be as specified in 5.1.1.3.3.

* 5.2.2.3 Fans together with accessories up to 65 pounds.- Each fan together with equipment weighing up to 65 pounds, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to type CF or SF, class domestic, variety SW, grade 200, and style optional of PPP-B-636, except that the minimum bursting strength of the fiberboard shall be 200 p.s.i. Contents shall be immobilized by means of fiberboard inserts made of type CF fiberboard as specified for the container. Each shipping container shall be closed in accordance with method II as specified in the appendix of the box specification.

* 5.2.2.3.1 When specified (see 6.2), the fiberboard shipping containers shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of that specification.

* 5.2.3 Level C.- Each fan, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers and packing shall comply with Uniform Freight Classification or National Motor Freight Classification as applicable.

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* 5.3 Palletization.- Fans packed as specified in shipping containers without sides, shall be palletized in accordance with load type 1 of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Special handling marking applicable to arrows and the words "THIS SIDE UP" shall be required on shipping containers with style B fans completely assembled.

6. NOTES

6.1 Intended use.- The fans covered by this specification are intended to be used to exhaust air primarily for light pressure application not exceeding 1/8 inch static pressure. These fans are not intended for use with duct work of any type.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size fan required (see 1.2).
(c) When a first article is required (see 3.1, 4.2 and 6.3).
(d) When fan characteristics are other than as specified herein (see 3.5.1).
(e) Voltage, frequency and phase characteristics of power source required (see 3.6.3).
(f) When V-belt protection is required and location of motor for class 2 fans (see 3.6.3.1).
(g) Power connection required (see 3.6.4).
(h) Fan accessories required (see 3.7).
    (1) Applicable NEMA-type of manual starter required; when type II motors are required; applicable class and group of hazardous location (see 3.7.1 and 3.7.1.2).
    (2) Type of reset required (see 3.7.1.1).
    (3) When other than a 3/4 inch conduit fitting hole is required and locations if more than 1 are required (see 3.7.1.2.2).

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(4) When installed fan height is less than 7 feet fan guard is required (see 3.7.2).
(5) When housing is required, specify type (penthouse or hood) (see 3.7.4), when blackout properties are required (see 3.7.4.2), and inside dimensions of housing base (see 3.7.4).
(6) When size of screen mesh is other than 1/2 inch square (see 3.7.5).
(i) When fungus control and moisture control is required (see 3.8).
(j) When electromagnetic compatibility is required (see 3.13).
(k) Finish requirements if other than specified (see 3.14).
(l) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
(m) Method of preservation required (see 5.1.1.2).
(n) When weather-resistant fiberboard shipping containers are required for level B packing (see 5.2.2.3.1).
(o) When palletization is not required (see 5.3).

* 6.2.1 Contract data requirements.- Any requirements for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

* 6.3 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be the initial production item. The first article should consist of one complete fan. The Contracting Officer should include specific instructions in all procurement instruments regarding arrangement for inspection and approval of the first article.

* 6.4 Standard air.- All reference made in this specification to air pressure, volumes, and velocities shall be considered as referring to standard air. Standard air shall be interpreted as dry air temperature of 70 F., a barometric pressure of 29.92 inches of mercury and a density of 0.075 pounds per cubic foot.

* 6.5 Additional application information.- In critical application where noise level is required to be held to a minimum because of the military mission, the following requirements and ordering data are necessary:

(1) The maximum loudness in sones allowed for the installed fan application.

(2) Actual application of fan use.

(3) Number of fans per application.

(4) Dimensions of room.

(5) Average sound absorption coefficient of room surface (if more than one coefficient applies, the area for each coefficient).

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(6) Room or background loudness in sones without fans as determined by AMCA 300 and 301.

(7) Type of sound test to be used per AMCA 300 for fan.

6.6 The margins of this specifcation have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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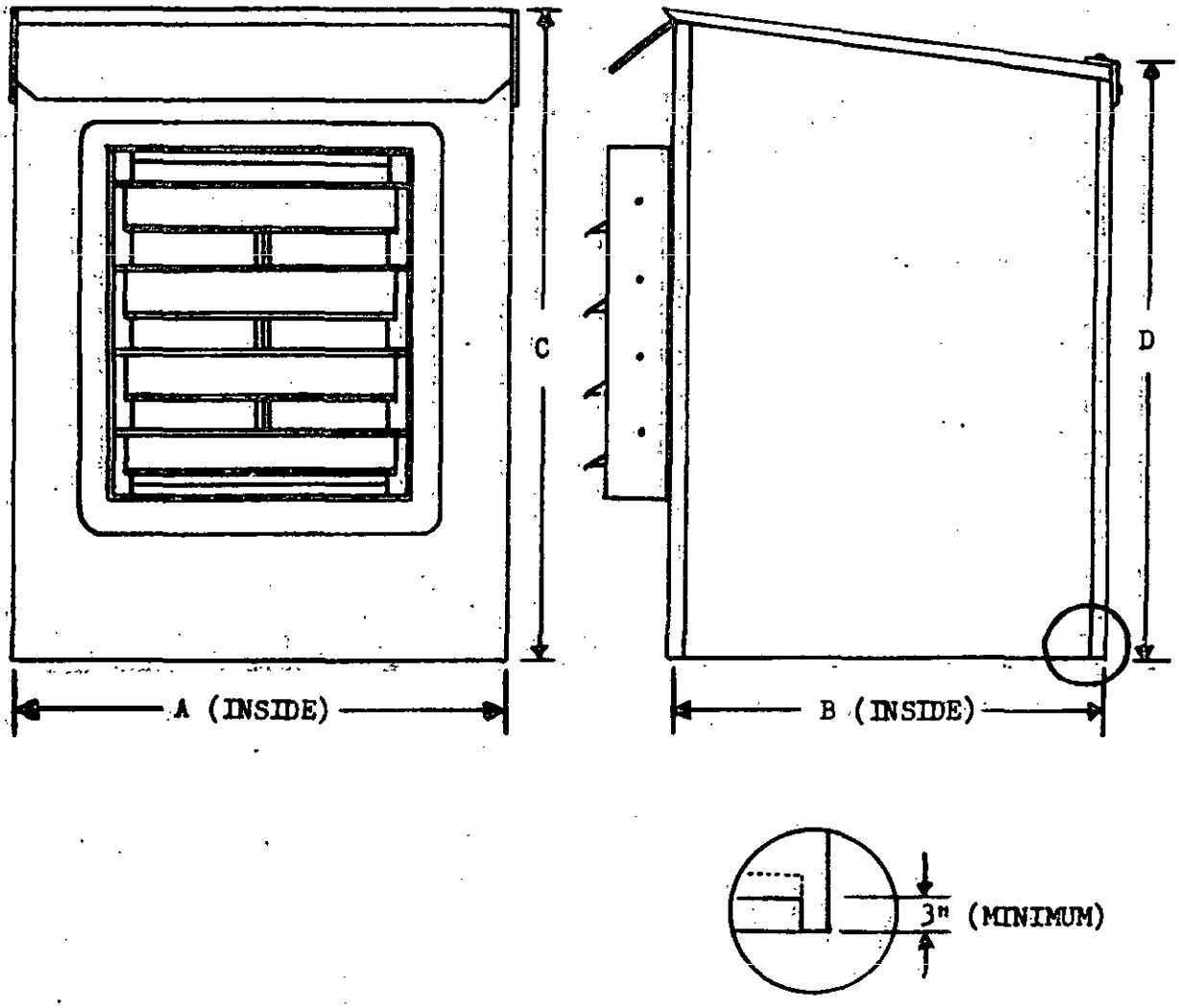
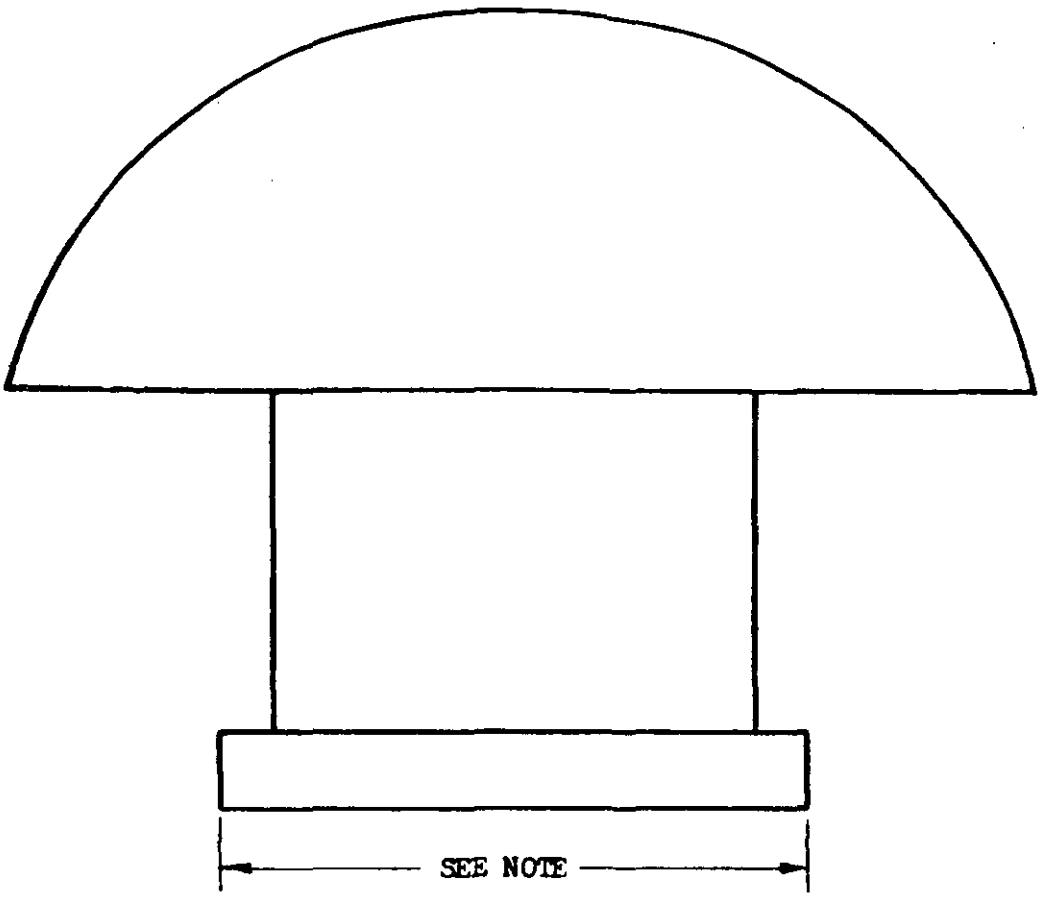


FIGURE 1. Penthouse Type Housing--Unit Mounting Detail
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Note: Inside dimensions of curbing shall be as specified. Curbing shall be square or rectangular.

FIGURE 2. Typical Power Roof Ventilator.
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